

Array Declaration & Initialization

Arrays store multiple values of the same type in a single variable. They are fixed-size once created.

```
String[] cars = {"Volvo", "BMW", "Ford"};  
int[] numbers = {10, 20, 30};
```

Accessing Elements

Java arrays use **Zero-Based Indexing**. The first element is at index 0.

Index	0	1	2
Value	Volvo	BMW	Ford

```
System.out.println(cars[0]); // Output: Volvo  
System.out.println(cars.length); // Output: 3
```

Iterating Through Arrays

Standard For-Loop: (Gives you access to the index `i`)

```
for (int i = 0; i < cars.length; i++) {  
    System.out.println(cars[i]);  
}
```

Enhanced For-Each: (Cleaner, no index needed)

```
for (String car : cars) {  
    System.out.println(car);  
}
```

Two-Dimensional Arrays

A multidimensional array is essentially an **Array of Arrays** (Rows and Columns).

```
int[][] matrix = { {1, 2, 3}, {4, 5, 6} };  
System.out.println(matrix[1][2]); // Output: 6 (Row 1, Col 2)
```

Pro Tip: Array Limitations

- Once created, you **cannot** change the size of an array.
- Use `System.arraycopy()` or `Arrays.copyOf()` to resize.
- For dynamic sizes, use `ArrayList` (advanced topic).